

# Question Paper

Exam Date & Time: 27-Nov-2025 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: ECE 2126/ECE\_2126 ANALOG CIRCUITS

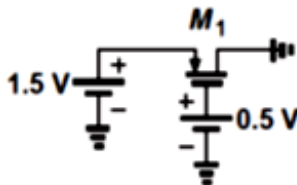
(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

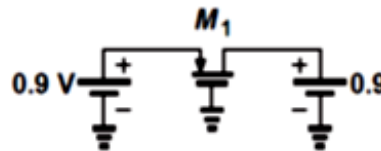
Duration: 180 mins.

Answer all the questions.

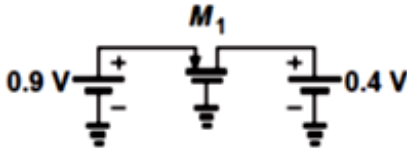
- 1A) Determine the region of operation of  $M_1$  in each circuit shown in the Figure below. Assume  $V_{TH} = 0.4V$ . (4)



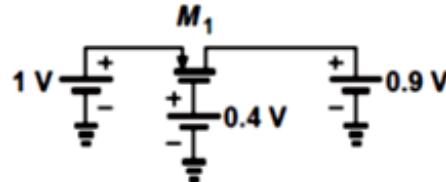
(a)



(b)



(c)



(d)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

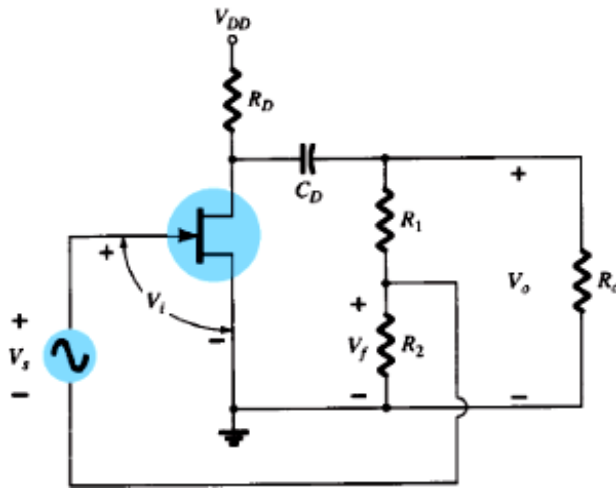
- 1B) An NMOS device operating in saturation with  $\lambda = 0$  must provide a transconductance of  $1/(50\Omega)$ . (3)  
Assume  $\mu_n C_{ox} = 100 \mu A/V^2$   
(a) Determine  $W/L$  if  $I_D = 0.5 \text{ mA}$ .  
(b) Determine  $W/L$  if  $V_{GS} - V_{TH} = 0.5V$ .  
(c) Determine  $I_D$  if  $V_{GS} - V_{TH} = 0.5V$ .

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Explain the basic block diagram of an operational amplifier with a neat diagram and plot its transfer characteristics. Also, explain the characteristics of an ideal op-amp with proper definitions. (3)

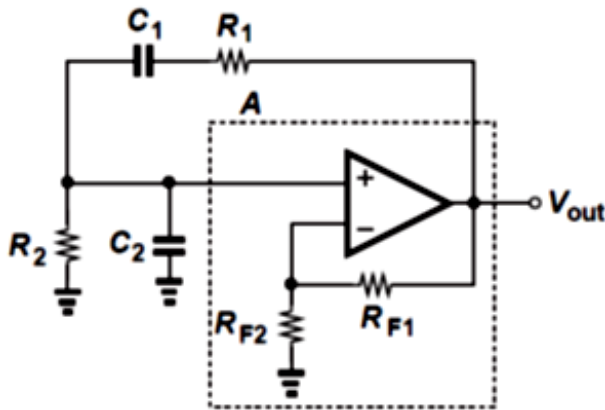
Course Outcome: "CO1" / Cognitive Level: "Remember"

- 2A) Calculate the gain with and without feedback for an FET amplifier as shown below for circuit values  $R_1 = 800 \text{ k}\Omega$ ,  $R_2 = 200 \Omega$ ,  $R_o = 40 \text{ k}\Omega$ ,  $R_D = 8 \text{ k}\Omega$ , and  $g_m = 5000 \mu S$ . (4)



Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2B) Design the Wein bridge oscillator of Figure shown below for a frequency of 10 MHz, assuming an ideal opamp. Assume  $C_1 = C_2 = 1$  nF. (3)



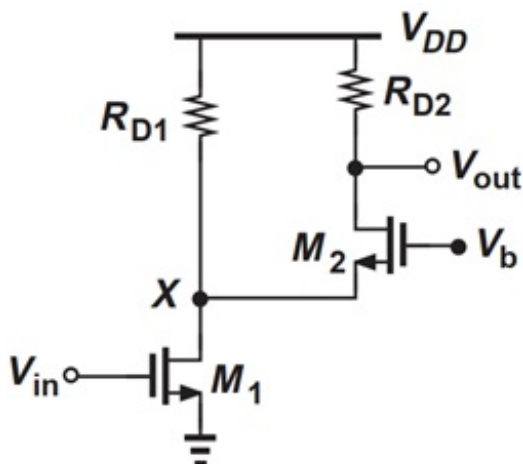
Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Explain the concept of Voltage-Current feedback with a neat diagram and derive the expressions for effective input impedance, output impedance, and gain. (3)

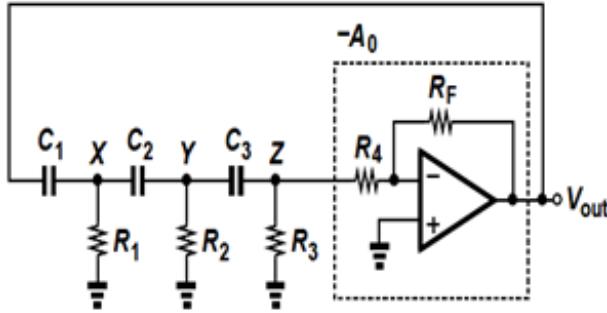
Course Outcome: "CO2" / Cognitive Level: "Understand"

- 3A) Consider the circuit of figure, where a common-source stage ( $M_1$  and  $R_{D1}$ ) is followed by a common-gate stage ( $M_2$  and  $R_{D2}$ ). (4)

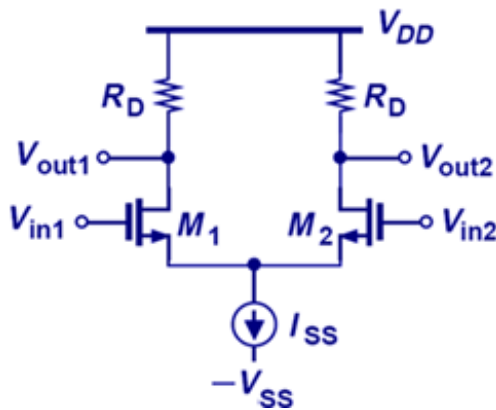
- (a) Writing  $V_{out}/V_{in} = (V_x/V_{in})(V_{out}/V_x)$  and assuming  $\lambda = 0$ , compute the overall voltage gain.  
 (b) Simplify the result obtained in (a) if  $R_{D1} \rightarrow \infty$ . Explain why this result is to be expected.



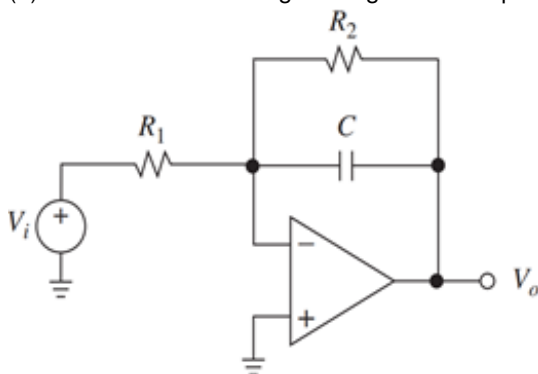
- 3B) Design the phase shift oscillator as shown in the figure for a frequency of 4 MHz with  $C_1 = C_2 = C_3 = 1\text{ nF}$ . Assume ideal op-amp behavior for the design. Find  $R_1 = R_2 = R_3$  and 5% feedback resistors to meet the oscillation gain requirement.



- 3C) A MOS differential pair has  $\mu_n C_{ox} = 100\text{ }\mu\text{A/V}^2$ ,  $W/L = 20/0.18$ ,  $V_{TH} = 0.5\text{ V}$ ,  $V_{DD} = V_{SS} = 1.5\text{ V}$ ,  $R_D = 2.5\text{ k}\Omega$ , and  $I_{SS} = 0.4\text{ mA}$ . The current source requires minimum  $0.4\text{ V}$  to operate properly.
- Calculate the equilibrium overdrive voltage and  $V_{GS}$  for each transistor.
  - For  $V_{CM} = 0\text{ V}$ , find  $V_S$ ,  $I_{D1}$ ,  $I_{D2}$ ,  $V_{D1}$ , and  $V_{D2}$ .
  - Determine the highest permitted value of  $V_{CM}$ .



- 4A) Design a first-order active low-pass filter (shown in figure) with gain using an op-amp that meets the following specifications: DC gain ( $H_0$ ) = 26 dB, 3-dB cutoff frequency ( $f_0$ ) = 2.5 kHz, Input resistance = 20 k $\Omega$
- Calculate the required component values ( $R_1$ ,  $R_2$ , and  $C$ ).
  - Determine the actual gain magnitude and phase angle at frequency  $f = 10\text{ kHz}$ .



- 4B) Using standard 5% resistor values, design a two-stage circuit that produces the output voltage  $V_o =$  (3)

$5V_1 + 10V_2 - 2V_3$ . Provide the complete circuit diagram and a detailed step-by-step solution justifying your component choices. The input impedance for the first stage inputs should be 22 k $\Omega$ .

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) What are the limitations of standard half-wave precision rectifier. Explain the improved/modified version of it with circuit diagram input and out waveform. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

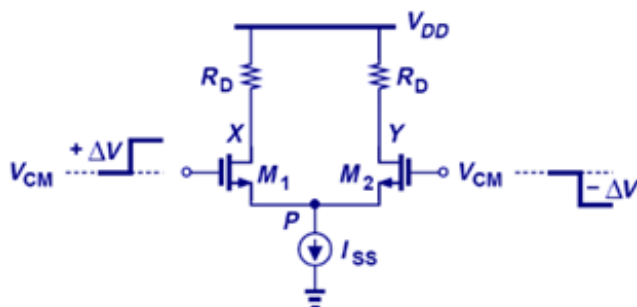
- 5A) Show the standard configuration for the astable multivibrator using the 555 timer with external components, that cannot achieve exactly 50% duty cycle. Evaluate this limitation by deriving the duty cycle expression. Explain why 50% duty cycle cannot be achieved and propose the circuit modifications to overcome this limitation. (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) Design a monostable multivibrator using an IC 555 timer that produces a positive output pulse with a duration of 200 ms. Select a capacitor value of 0.2  $\mu$ F, and calculate the required resistance. Draw the final (a) circuit diagram, (b) voltage across the capacitor, (c) trigger input and (d) output waveform. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) (3)



For the MOS differential pair shown in the figure, a common-mode input  $V_{CM}$  i.e., a small differential input signal  $\Delta V$  is applied such that  $V_{in1} = V_{CM} + \Delta V$  and  $V_{in2} = V_{CM} - \Delta V$ . Derive an expression for the small-signal differential voltage gain,  $A_v = (\Delta V_x - \Delta V_y) / (2\Delta V)$ . Show that this gain is similar to that of a common-source stage.

Course Outcome: "CO3" / Cognitive Level: "Application"

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| Event Name | Subject |
|------------|---------|
|------------|---------|

058-END SEMSTER-ODD-2025 ECE\_2126:Analog Circuits

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 10        |
| 2            | 2              | CO 2        | 17        |
| 3            | 3              | CO 3        | 6         |
| 4            | 4              | CO 4        | 10        |
| 5            | 5              | CO 5        | 7         |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
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058-END SEMSTER-ODD-2025 ECE\_2126:Analog Circuits

| No of |
|-------|
|-------|

| # | Level       | Questions | Marks (%) |
|---|-------------|-----------|-----------|
| 1 | Remember    | 1         | 3(6%)     |
| 2 | Understand  | 2         | 6(12%)    |
| 3 | Application | 10        | 33(66%)   |
| 4 | Analysis    | 2         | 8(16%)    |
| 5 | Create      | 0         | 0(0%)     |
| 6 | Evaluate    | 0         | 0(0%)     |
|   | Total       | 15        | 50 (100%) |